PRODUCT REQUIREMENTS DOCUMENT

Aura

Where Music Becomes Visual — A Generative Art Experience Platform

Version	1.0 — Draft
Date	March 2026
Status	For Review
Product Type	Two-Sided Platform — Artist Studio + Listener Experience
Art Engine	Real-time generative AI (Stable Diffusion / Flux) + curated override
Response Layers	Mood/emotion + audio frequency/beat + lyric/narrative (all three)

1. Product Overview

Aura is a generative visual music experience platform that transforms any album or song into a living, breathing visual world — displayed on a second screen, smart TV, or companion display while the music plays.

It is a two-sided platform. Artists use Aura Studio to define the visual language of their music: the color palette, art style, emotional arc, and key moments they want visualized. Listeners experience Aura Player — a generative art display that responds to the music in real time, shaped by both the artist's vision and three live signal layers: the emotional character of the song, the raw audio signal, and the lyrical narrative.

The art is not a static video. It is not a visualizer bouncing to the beat. It is a continuously evolving piece of generative art that understands the music — its mood, its rhythm, its story — and translates all three into a visual experience that no two listeners see the same way twice.

The Core Vision

When Kendrick Lamar drops a new album, a visual world should come with it. Not a music video. Not album art. A generative universe that opens every time you press play — shaped by his vision, alive with every bassline, shifting with every lyrical turn. Aura is the infrastructure for that. For every artist. For every album.

2. Problem Statement

2.1 Music Is Still Experienced Eyes-Closed

Streaming solved music discovery and access. It did not solve the visual dimension of the listening experience. When you play an album on Spotify, Apple Music, or Tidal, you see a static image — the album cover — and nothing else. The richest sensory component of music, the visual world it evokes, is left entirely to the listener's imagination.

Artists invest deeply in visual identity — album art, music videos, stage design, merch. But none of that visual language follows the listener into the room where they actually listen. The album plays. The room stays blank.

2.2 Existing Solutions Are Not the Answer

Category	What Exists	Why It Falls Short
Spotify Canvas	Short looping video clips on the Spotify mobile app (max 8 seconds)	Static loop. No reactivity to audio. No generative depth. Not displayed on a separate screen.
Music visualizers	Winamp-style frequency bars, iTunes visualizer, Spotify's own basic visualizer	Generic. Not music-specific. Zero artistic intention from the artist. All songs look the same. No lyrical or emotional response.
Music videos	Pre-produced narrative videos for specific singles	Expensive. Cover 1% of a catalog. Linear — doesn't react to the audio. Designed for one watch, not the 50th listen.
Ambient art displays	Digital frames (Meural, Samsung Frame), NFT display platforms	Static images. No connection to what is playing. No reactivity. No artist music-specific curation.
Live VJ / visual performance	Real-time visuals at concerts	Designed for live performance, not home listening. Not accessible per song or album.

2.3 What the Market Is Missing

- A visual layer that travels with the music — not the stage, not the video, but the everyday listening session
- Generative art that responds to all three dimensions of music simultaneously: its emotional character, its physical audio signal, and its lyrical content
- Artist authorship over the visual language — not a generic AI hallucination, but a visual world shaped by the artist's own aesthetic choices
- A display-first design — built for the TV, the digital frame, the tablet on the bookshelf, not just a phone screen corner

3. User Personas

Persona	Who They Are	What They Want	How Aura Serves Them
The Artist	Musician, producer, or label releasing an album. Has a strong visual identity and aesthetic references. May work with a visual director or art director.	A visual world that extends their album's identity into every listening session — shaped by their vision, not by an algorithm they have no control over.	Aura Studio: define a Style Brief (palette, art style, mood arc, reference images), annotate key moments in songs for specific visual triggers, review and approve AI-generated previews, curate specific frames for specific songs.
The Deep Listener	Audiophile, music enthusiast. Listens to full albums. Has a dedicated listening setup — good speakers, maybe a TV or monitor near the listening chair.	A visual experience worthy of the music — something that deepens immersion rather than distracting from it. Something that makes the 50th listen of an album feel new.	Aura Player on their TV or second screen. The art shifts with every song, reacts to every drop, evolves with every lyrical movement. Each session is unique.
The Ambient Listener	Uses music as a background for work, dinner, or relaxation. Has a digital frame, smart TV, or tablet in the room.	Something beautiful on screen that responds to the music passively — doesn't need to be watched, just felt.	Aura's ambient mode: slower generative evolution, softer audio reactivity, more painterly and less kinetic.
The Label / Visual Director	Art director, creative team, or label managing an artist's visual identity. May be setting up Aura on behalf of the artist.	Control. The ability to define the visual language, approve outputs, and ensure every generated frame is consistent with the artist's brand.	Aura Studio with team access: multiple collaborators, approval workflow, asset library management, style brief version control.

4. The Three Response Layers

Aura responds to music across three simultaneous layers. Each layer feeds the Visual Generation Agent independently and their signals are blended according to artist-defined weights. The default blend is equal, but artists can emphasize any layer — an artist whose work is primarily lyric-driven might weight narrative 60%, mood 30%, audio 10%.

Layer 1 — Mood / Emotion

What the song feels like emotionally — its valence (positive/negative), energy (high/low), tension, release, intimacy, intensity. This is the slowest-changing layer. It sets the overall palette, texture, and atmospheric character of the visual.

- Sourced from: Spotify Audio Features API (valence, energy, danceability, acousticness, instrumentalness, mode), Claude-based lyric sentiment analysis, and artist-defined emotional metadata in the Style Brief
- Changes at: song level and section level (verse/chorus/bridge transitions often shift emotional register)
- Visual expression: dominant color palette, texture warmth or coldness, density of visual elements, presence of light vs. shadow, sharpness vs. softness
- Example: A low-energy, high-valence song (gentle joy) produces warm, soft, golden-toned imagery. A high-energy, low-valence song (angry or aggressive) produces sharp, dark, high-contrast imagery with kinetic movement.

Layer 2 — Audio Signal

What the music physically does — its beat, tempo, frequency spectrum, amplitude, and dynamic range. This is the fastest-changing layer. It drives moment-to-moment animation and reactive motion in the visual.

- Sourced from: Real-time audio analysis via Web Audio API (browser) or native audio buffer analysis (mobile/desktop). Beat detection, BPM tracking, frequency band separation (sub-bass, bass, mid, high), amplitude envelope.
- Changes at: beat level, bar level, dynamic peaks and drops
- Visual expression: pulse and scale on beats, particle system density tracking amplitude, color saturation spiking on transients, motion speed matching BPM, bass frequencies driving low-frequency visual 'weight' (heavy dark forms), high frequencies driving brightness and lightness
- Example: At a 808 kick, the dominant visual element pulses outward. At a snare crack, a flash of light fractures through the image. During a breakdown with no drums, the image slows to near-stillness.

Layer 3 — Lyric / Narrative

What the song says — its imagery, themes, references, and emotional narrative arc. This is the most conceptually rich layer. It drives the subject matter and symbolic content of what is generated.

- Sourced from: Lyric transcription (Genius API for known tracks, Whisper for unrecognized), Claude-based semantic analysis for imagery, metaphor extraction, and scene identification
- Changes at: lyric line level — the agent identifies concrete images and scenes embedded in the lyrics and translates them into visual prompts
- Visual expression: specific subject matter appears — places, objects, moods, and archetypes derived from the lyrics. 'Standing on the mountain watching cities burn' produces dramatically different imagery than 'walking through the rain on a Tuesday'
- Artist control: artists can annotate specific lyric lines with visual overrides — 'at this line, I want THIS image' — and those annotations always override the AI-generated interpretation

How the Three Layers Blend

Think of it as three painters working simultaneously on the same canvas. Layer 1 (Mood) chooses the palette and overall atmosphere. Layer 2 (Audio) animates — it decides what moves, how fast, and with what intensity. Layer 3 (Lyrics) chooses what things appear — the subjects, symbols, and scenes. The artist's style brief constrains all three within their aesthetic universe. No layer can override the artist's visual identity.

5. Agentic Architecture

5.1 System Overview

Aura runs five agents. Four operate continuously during playback — they are the real-time intelligence that drives the visual. The fifth operates in the Aura Studio context — it learns the artist's visual language and maintains it across every generated frame.

The pipeline has two tracks running in parallel:

- Pre-generation track (runs once per song, before or during first play): Lyric Agent + Mood Agent analyze the song and produce a Song Visual Blueprint — a structured set of scene prompts, palette guidance, and key moment triggers ready for the Visual Generation Agent.
- Real-time track (runs continuously during playback): Audio Analysis Agent feeds live signal data to the Visual Generation Agent every 2-4 seconds, triggering animation updates and periodic full frame regenerations.

The Artist Style Agent runs in the background on both tracks, injecting the artist's Style Brief as a persistent constraint on every Visual Generation Agent call. It ensures that a frame generated at 2:34 of track 7 looks like it belongs to the same world as a frame generated at 0:15 of track 1.

5.2 Agent Specifications

Agent 01 Audio Analysis Agent — Parse the live audio signal into structured visual parameters

Trigger: Continuous — fires every 100ms during playback, aggregates into 2-second update packets

Inputs: Live audio buffer (from Web Audio API or platform audio stream), current BPM, current playback position, song's audio feature baseline (from Spotify API or pre-analysis)

Outputs: Audio Parameter Object updated every 2 seconds: BPM, current beat phase (0-1), amplitude (overall + per frequency band: sub-bass / bass / mid / high), dynamic range, detected section change flag (verse/chorus/bridge), energy delta from baseline

Tools / APIs: Web Audio API (browser), platform audio hook (desktop/mobile), Spotify Audio Analysis API (pre-computed beat grids, section markers), librosa (Python audio analysis for pre-processing), Essentia.js (client-side real-time analysis)

Priority: P0 — Core. The heartbeat of the real-time visual system.

Agent 02 Lyric & Narrative Agent — Extract visual scenes and imagery from lyrics

Trigger: Runs once per song on first play (or on song load in pre-generation mode). Re-runs if lyrics change or artist annotations are updated.

Inputs: Full lyric text (from Genius API or Whisper transcription), song metadata (title, artist, album, genre), artist's Style Brief

Outputs: Song Narrative Map: timestamped lyric segments matched to playback position, per-segment visual prompt (concrete imagery distilled from lyric content), dominant metaphor and theme tags, intensity arc across the song, flagged key moments (emotional peaks, repeated hooks, dramatic lines), artist annotation overrides applied

Tools / APIs: Genius API (lyrics + annotations), Whisper API (transcription for unrecognized tracks), Claude claude-sonnet-4-6 (semantic analysis, imagery extraction, scene prompt generation), LRC timestamp format parser (for synced lyrics)

Priority: P0 — Core. Pre-generates the narrative layer so it is ready before the song plays.

Agent 03 Mood & Emotion Agent — Translate the song's emotional character into a visual atmosphere

Trigger: Runs once per song on load. Updates at detected section transitions (verse to chorus, etc.) during playback.

Inputs: Spotify Audio Features (valence, energy, danceability, acousticness, instrumentalness, tempo, mode, key), lyric sentiment analysis from Lyric Agent, artist-defined mood metadata from Style Brief, section markers from Audio Analysis Agent

Outputs: Mood Parameter Object per song and per section: emotional quadrant (valence x energy mapped to 2D space), dominant emotional tone label (e.g., 'triumphant', 'melancholic', 'anxious', 'euphoric'), palette suggestion (warm/cool, saturated/muted, light/dark), texture character (sharp, soft, fluid, fractured), recommended generative style bias

Tools / APIs: Spotify Web API (/audio-features), Claude claude-sonnet-4-6 (emotional analysis, palette translation), internal emotion-to-visual mapping rules (artist-configurable)

Priority: P0 — Core.

Agent 04 Artist Style Agent — Maintain the artist's visual identity across every generated frame

Trigger: Runs as a persistent context injection on every Visual Generation Agent call. Updated when artist modifies their Style Brief in Aura Studio.

Inputs: Artist's Style Brief (color palette, art style, reference images, visual exclusions, key symbols, aesthetic references), album-level visual theme, approved frame history for this artist

Outputs: Style Constraint Object injected into every image generation prompt: locked palette tokens (HEX values the image must respect), style descriptor tags (e.g., 'oil painting texture', 'geometric abstraction', 'dark surrealism', 'afrofuturist'), negative prompt tokens (visual elements explicitly excluded by artist), continuity anchors (recurring visual motifs that should appear across the album)

Tools / APIs: Claude claude-sonnet-4-6 (style brief interpretation, prompt engineering), internal Style Brief data model, artist asset library (uploaded reference images, color swatches, example art)

Priority: P0 — Core. The artist's fingerprint on every generated frame.

Agent 05 Visual Generation Agent — Generate and animate the visual output in real time

Trigger: Triggered every 4-8 seconds for full frame regeneration. Triggered every 100-200ms for animation parameter updates (no full regen — just motion/transform updates on existing frame).

Inputs: Audio Parameter Object (from Agent 01), current Song Narrative Map segment (from Agent 02), Mood Parameter Object (from Agent 03), Style Constraint Object (from Agent 04), current playback timestamp, last generated frame (for continuity), display mode (immersive / ambient / minimal)

Outputs: Generated visual frame (1920x1080 or 4K depending on display), animation parameter update (scale, rotation, blur, color shift values applied to current frame between full regens), frame transition instruction (cut / dissolve / morph), display metadata (current song, timestamp, agent blend weights for this frame)

Tools / APIs: Stable Diffusion XL / Flux.1 (image generation via Replicate API or self-hosted), ControlNet (for frame-to-frame continuity and motion guidance), AUTOMATIC1111 API or ComfyUI (pipeline orchestration), Three.js / WebGL (client-side real-time animation of generated frames), CSS transform animations (lightweight motion between regens)

Priority: P0 — Core. Most computationally intensive agent. Pre-generation for known tracks; real-time for unknowns.

6. Aura Studio — Artist Side

Aura Studio is where artists and their teams define the visual language of their music. It is a web-based creative tool — not a complex DAW, not a design suite. It is a structured brief-building environment that feeds directly into the generative pipeline.

6.1 Style Brief Builder

- System MUST allow artists to define a Style Brief per album and per individual track, with album-level settings inherited by tracks unless overridden

- Style Brief components: color palette (up to 8 HEX values, labeled by emotion/use), art style descriptors (selected from a curated taxonomy + free text), reference images (uploaded or linked, tagged as 'vibe reference' / 'avoid this look'), visual exclusions (things the artist explicitly does not want generated — people, specific imagery, etc.), dominant symbols or motifs (a recurring visual element to weave through the album)
- System MUST include a reference image upload: artist can upload 3-10 images that represent their visual world — artwork they love, photography references, their own album art, tour visuals
- System MUST support layer weighting: artist sets the blend between the three response layers (Mood / Audio / Lyrics) per album — default 33/33/33
- System SHOULD include a starter template library: 20+ curated style archetypes (Dark Surrealism, Afrofuturism, Lo-fi Nostalgia, Abstract Geometry, Impressionist Landscapes, Cyberpunk Neon, etc.) that artists can use as a starting point and customize

6.2 Song Annotation

- System MUST allow artists to annotate specific moments in a song with visual instructions: 'at 1:32 show isolation, blue tones' or 'when the hook hits, everything goes white'
- System MUST display a scrollable waveform + lyric view where artists can add timestamped annotations by clicking on the waveform
- System MUST support image pin annotations: artist can upload a specific image and pin it to a timestamp — when playback reaches that moment, that image (or a close generative variation of it) is displayed
- System MUST support lyric-line annotations: specific lyric lines can be overridden with a visual instruction that supersedes the AI interpretation

6.3 Preview & Approval

- System MUST generate a preview render for any song in the artist's catalog — a 60-second looping preview showing how the generative art looks for that track
- System MUST allow artists to regenerate any preview with different settings and compare side by side
- System MUST support a live preview mode: artist presses play on their song, the generative display runs in real time in the Studio interface so they experience exactly what the listener will
- System SHOULD support a 'lock frame' feature: if the artist sees a generated frame they love during preview, they can lock it to that timestamp — that frame (or a variation) will always appear at that moment

6.4 Team Access

- System MUST support multi-seat access: artist, manager, visual director, and label team can all access the same Studio workspace

- System MUST include a simple approval workflow: visual director proposes a Style Brief, artist approves before it goes live to listeners
- System SHOULD include a comment layer on the preview: team members can leave timestamped visual comments on the preview render

7. Aura Player — Listener Side

Aura Player is the listener's experience. It is designed for a second screen — a TV, a digital frame, a tablet, or a large monitor — that lives in the same room as the speaker. It is not designed as a primary interface. It is designed to be glanced at, absorbed, and occasionally stared into.

7.1 Display Modes

- Immersive mode — full screen, high-detail generative art, full audio reactivity, designed for dedicated listening sessions. The primary experience.
- Ambient mode — softer generation, slower evolution, reduced audio reactivity. Designed for background listening. More painterly, less kinetic. Screen never goes fully dark.
- Minimal mode — clean framing, generous negative space, muted palette override. Designed for focus work or dining settings where visual distraction should be low.
- Artist Vision mode — displays only artist-curated frames and approved generative outputs. AI creativity is constrained to the artist's tightest specifications. The most intentional experience.

7.2 Second Screen Integration

- System MUST support casting to any Chromecast, AirPlay, or Miracast-compatible display with one tap
- System MUST support Samsung SmartThings and LG ThinQ TV integration — Aura Player available as a native TV app
- System MUST support ambient display on Android tablets and iPads (always-on mode with battery management)
- System SHOULD support digital frame integrations: Meural, Canvia, and Samsung Frame TV as dedicated display targets
- System SHOULD support WebSocket-based local network sync: phone controls playback, TV displays the visual — no cloud round-trip for display updates

7.3 Music Source Integration

- System MUST integrate with Spotify (OAuth) — Aura reads what is playing and responds in real time via Spotify Connect API and playback state polling
- System MUST integrate with Apple Music (MusicKit JS) — same read-and-respond architecture

- System MUST support local file playback via the Aura desktop app (Electron) — for audiophiles with local FLAC/ALAC libraries
- System SHOULD integrate with Tidal (for lossless and MQA listeners who represent a high-engagement segment)
- System SHOULD integrate with YouTube Music
- System MUST handle 'unknown track' gracefully — when a song has no artist Style Brief and no Genius lyric data, Aura falls back to a fully AI-driven generative mode using whatever audio signal and mood analysis is available

7.4 Listener Personalization

- System SHOULD allow listeners to set their own default style preferences when no artist brief exists — 'I prefer abstract and dark' vs 'I prefer painterly and warm'
- System SHOULD offer a 'style slider' during playback: listener can shift the visual in real time between the artist's vision and their own preference
- System SHOULD support saving a specific generated frame as a still — listener can save any frame they love to their personal gallery
- System MAY allow listeners to share a generated frame with the song metadata attached — social sharing of Aura frames as a discovery mechanism for the platform

8. The Visual Continuity Problem & Solution

The hardest technical problem in Aura is not image generation — it is visual continuity. If each frame is generated independently, the result is a jarring slideshow of unrelated images. A listening experience that constantly cuts to a completely different scene every 4-8 seconds breaks immersion rather than creating it.

Aura solves visual continuity through four mechanisms:

1. Anchor-Based Generation

Every new frame generation uses the previous frame as a ControlNet conditioning input — the new frame must maintain structural and compositional similarity to the prior frame, evolving rather than replacing it.

2. Prompt Drift

The text prompt feeding the image generator changes gradually over time, not abruptly. When the lyric layer shifts from 'isolation in a dark room' to 'stepping into light,' the prompt interpolates between the two states over 30-60 seconds rather than cutting instantly.

3. Animation Layer

Between full frame regenerations (every 4-8 seconds), the current frame is animated using CSS transforms, WebGL shaders, and Three.js particle systems — slow zooms, subtle color shifts, particle drift, light bloom. This fills the gap between generations with motion that feels intentional, not like a loading state.

4. Pre-Generation for Known Tracks

For any track where the artist has set a Style Brief and lyric data is available, Aura pre-generates the full visual arc of the song before the listener presses play. When playback begins, the system is already cycling through a pre-rendered generative sequence — real-time reactive animation is overlaid on the pre-generated frames, but the fundamental visual journey of the song is ready instantly.

Continuity Mechanism	What It Solves	When It Runs
ControlNet anchor	Prevents total scene discontinuity between frames	Every full frame regeneration
Prompt drift	Prevents abrupt subject/palette changes during narrative transitions	Section transitions, key lyric shifts
Animation layer	Fills the gap between full regenerations with living motion	Continuously between full regens (every 100-200ms)
Pre-generation	Eliminates loading latency and enables the full 3-layer arc to be planned as a coherent journey	On song load for known tracks with Style Brief

9. Non-Functional Requirements

Category	Requirement
Latency — real-time	Animation layer updates: < 100ms. Full frame regeneration: < 4 seconds (server-side generation) or < 8 seconds (on-device). Pre-generated frames: < 200ms transition latency.
Display quality	Output resolution: 1920x1080 minimum. 4K (3840x2160) for Pro tier. HDR metadata support for compatible displays. Frame rate: 60fps for animation layer over generated still frames.
Audio sync precision	Beat-reactive visual events must sync within ±50ms of the detected beat. Section change visual response within ±200ms of detected section boundary.
Offline / degraded mode	If generation server is unavailable, Aura falls back to a curated library of pre-rendered ambient visual loops matched to the song's mood category. Never a blank screen.
Privacy	Listening history used only to improve personalized style preferences —

	never sold or shared. Spotify and Apple Music tokens stored with user consent, revocable at any time.
Battery / performance	Ambient mode on tablet must not drain more than 10% battery per hour. GPU usage cap on mobile to prevent device overheating during extended sessions.
Accessibility	All UI controls accessible via keyboard and screen reader. Visual display modes should be pausable. Seizure risk: rapid flash effects must be suppressible via accessibility settings.

10. Technical Architecture

10.1 Stack

- Image generation: Stable Diffusion XL + ControlNet via Replicate API (cloud) or self-hosted ComfyUI pipeline (enterprise). Flux.1 as alternative for higher quality at higher cost.
- Client-side animation: Three.js + WebGL (particle systems, shader effects, light bloom). CSS transforms for lightweight zoom/pan between full regens.
- Audio analysis: Web Audio API (browser real-time). Essentia.js (client-side beat detection). Spotify Audio Analysis API (pre-computed beat grids and sections for Spotify tracks).
- Lyric intelligence: Genius API (known tracks). Whisper API (unknown tracks). Claude claude-sonnet-4-6 (semantic analysis, imagery extraction, scene prompt generation).
- Mood analysis: Spotify Audio Features API. Claude (sentiment analysis of lyrics). Internal emotion-to-visual mapping model.
- Backend: Python / FastAPI (agent orchestration, generation pipeline, API gateway).
- Task queue: Celery + Redis (pre-generation jobs, async frame rendering, fan-out to multiple active listeners).
- Database: PostgreSQL (artist profiles, style briefs, annotations, listener preferences). S3 (pre-generated frame libraries, artist uploaded assets).
- Realtime sync: WebSockets (Aura Player display synced to phone controller without cloud round-trip).
- Frontend: Next.js (Studio web app, Player web app). Electron (desktop Player for local file support). React Native (mobile controller app).
- TV / display: Android TV app (React Native + TV SDK). Chromecast receiver app. AirPlay mirroring (no custom receiver needed).

10.2 Generation Pipeline Flow

Step	What Happens	Runs When
1. Song Load	Lyric Agent fetches and maps lyrics. Mood Agent	When user presses play or

	calculates audio features. Artist Style Agent loads Style Brief. Pre-generation job queued.	adds song to queue
2. Pre-Generation	Visual Generation Agent renders 10-15 key frames for the song's arc based on Lyric Map + Mood Profile + Style Brief. Stored in S3.	Background, before or during first play
3. Playback Start	Pre-generated frames begin cycling. Animation layer activates. Audio Analysis Agent begins real-time processing.	On playback begin
4. Real-Time Beat Response	Audio Analysis Agent emits beat events. Animation layer responds with scale/pulse/color effects on current frame.	Every detected beat
5. Section Transition	Mood Agent re-evaluates for new section. Prompt drift begins toward new narrative/mood prompt. New full frame regen triggered.	On detected section change (verse/chorus/bridge)
6. Full Frame Regeneration	Visual Generation Agent generates a new frame with ControlNet anchor from current frame. Smooth transition applied.	Every 4-8 seconds or on narrative trigger
7. Key Moment Trigger	Artist-annotated timestamp reached. Pinned frame or locked visual instruction overrides generative flow.	At annotated timestamps
8. Song End	Final frame held, slows to near-stillness. Transition to next song's visual world via dissolve.	On song completion

11. Business Model

11.1 Two Revenue Tracks

Aura has revenue on both sides of the platform — listener subscriptions and artist/label studio fees.

Listener Side

Tier	Price	What's Included
Free	$0/mo	Artist Vision mode only (pre-curated frames where available). No AI generation for uncurated tracks. Standard definition. No second screen casting.
Listen	$6/mo	Full generative AI for all tracks. All three response layers. Second screen casting. 1080p. Ambient mode. Personal style preferences.
Listen Pro	$12/mo	Everything in Listen. 4K output. Frame saving gallery. HDR display support. Offline mode (pre-downloaded visual arcs for saved albums). Priority generation queue.

Artist / Label Side

Tier	Price	What's Included
Artist Free	$0/mo	1 album Style Brief. Basic style controls. Preview generation (5 renders/month). No team access.
Artist Studio	$29/mo	Unlimited albums. Full Style Brief toolkit. Unlimited preview renders. Song annotation. 2 team seats. Priority pre-generation for listeners.
Label	$149/mo	Unlimited artists and albums under one workspace. Full annotation suite. 10 team seats. API access for direct integration with artist release workflows. Dedicated support.

11.2 Additional Revenue

- Generation compute pass-through: free tier listeners pay per-generation credits for AI-generated frames beyond the monthly included volume
- Frame NFT / print: listeners can mint a generated frame as a limited edition collectible or order a print — revenue share with the artist whose music generated it
- Sync licensing integration: music supervisors and brands can license specific generative visual styles from artists for use in commercial contexts

12. User Stories

Artist

- As an artist releasing an album, I want to define a color palette and 5 reference images so that every generated frame feels like it belongs to the world I built.
- As an artist, I want to pin a specific image to the bridge of my most emotional track so that listeners always see exactly what I envisioned at that moment.
- As an artist, I want to preview the full generative visual arc of my album before it goes live so I can approve it the same way I approve album art.
- As a visual director working with an artist, I want to share a Style Brief draft with the artist and get their approval before the visuals go live to listeners.

Listener

- As a listener, I want to cast the Aura visual to my TV while the music plays from my phone so I can sit back and experience both simultaneously.
- As a deep listener, I want the visual to react to the beat so that drop in the third track hits me visually the same way it hits me sonically.

- As an ambient listener, I want the art to evolve slowly and beautifully in the background while I work — something that's alive but not distracting.
- As a listener, I want to save the specific frame that appeared during my favorite lyric so I can keep it.

13. Out of Scope — V1

- Live concert mode — real-time Aura visuals on stage or at events. The architecture is similar but the latency and display requirements are fundamentally different. V2 with a hardware/venue partnership track.
- Artist uploads of their own generative art or animated video — V1 uses Aura's generation pipeline. Custom artist-supplied animation is a V2 feature.
- Music production integration — reacting to a DAW's audio output during production. A compelling future use case for producers, not V1.
- Social/multiplayer listening — shared visual rooms where multiple listeners see and react to the same generative art together. Strong V2 concept.
- Lyric display overlay — showing the lyrics on screen over the visual. Technically simple but a design decision deferred to V2 based on user research.
- Podcast and spoken word support — the audio and lyric layers work for music; applying them to podcasts requires different emotional modeling. V3.

14. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Visual continuity breaks — jarring frame jumps ruin immersion	High (early)	Critical	ControlNet anchor on every generation. Pre-generation for known tracks. Animation layer fills gaps. Aggressive QA on continuity across all genre types.
Generation latency too slow for real-time feel	Medium	High	Pre-generation eliminates latency for known tracks. Animation layer keeps the screen alive during generation. Server-side generation with CDN caching of pre-generated frames. Local generation option for Pro users with capable GPUs.
Music streaming API access changes (Spotify revokes real-time access)	Medium	High	Local file playback as a fallback. Multi-platform strategy (Spotify + Apple Music + Tidal + local). Negotiate data partnership with streaming platforms — Aura is a value-add for their platforms, not a competitor.
Artist adoption low — labels and artists don't	Medium	High	Free Artist Studio tier removes financial barrier. Build initial catalog through independent artists. Show data: listeners with artist Aura briefs listen 35% longer per

prioritize this			session (internal beta metric to validate).
Generated art produces offensive or brand-inconsistent imagery	Low	Critical	Strong negative prompt defaults. Artist-controlled exclusion lists. Content moderation filter on all outputs before display. Human review queue for artist-flagged outputs.
Beat detection inaccurate for complex rhythms (polyrhythm, jazz, electronic)	Medium	Medium	Essentia.js + Spotify beat grids as dual source. Graceful degradation: if beat detection confidence is low, fall back to amplitude envelope for audio reactivity. Genre-specific beat detection tuning.

15. Open Questions

- Streaming API real-time audio access: Spotify's API provides playback state (what is playing, at what position) but not the actual audio stream for privacy/licensing reasons. How do we get the audio signal for the Audio Analysis Agent — from the device's audio output, from a companion microphone, or from pre-analyzed beat grids only?
- Artist rights to generated visuals: If Aura generates a frame during an artist's song using their Style Brief, who owns that frame? The artist, the listener (if they save it), or Aura? Legal framework needed before launch.
- Copyright exposure in reference images: Artists upload reference images for style inspiration — does using those images as ControlNet conditioning or style transfer inputs create copyright liability?
- Continuity as a creative choice: Some artists may want jarring, discontinuous visuals as an aesthetic choice. How do we expose a continuity dial that lets artists opt into fragmentation?
- On-device generation: Apple Silicon and high-end Android chips can now run Stable Diffusion locally. Is a local generation mode worth the development investment for the premium tier, or is cloud generation fast enough to make it unnecessary?
- Platform strategy: Is Aura a standalone app or does it seek to be embedded inside Spotify, Apple Music, or Tidal as a licensed feature? The embedded path is faster distribution but surrenders control. The standalone path is harder but defensible.

The Cultural Moment

Music and visual art have always been intertwined — album covers, music videos, stage design. But the everyday listening experience, the 45 minutes someone spends with an album on a Tuesday evening, has always been visually empty. Aura fills that space not with a screen saver or a video loop, but with a living visual world that the artist intended and the music generates. That is a new medium. And new mediums do not get built by iteration on existing ones.

This document is a working draft. Technical stack choices are directional and subject to revision during engineering scoping. Streaming API capabilities are subject to platform terms of service changes.